

Parnassus: Neural Fast Simulation and Reconstruction for High Energy Physics

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Abstract

We present the public software release of PARNASSUS, a pure-Python, GPU-compatible framework for fast detector simulation and reconstruction in High Energy Physics. Parnassus replaces the computationally expensive GEANT4 simulation and reconstruction chain with two interchangeable generator backends: a **neural backend** based on conditional flow matching models that map truth-level particles directly to detector-level particle-flow objects, and a **parametric backend** (TORCHDELPHES) that re-implements the DELPHES fast simulation chain in pure PyTorch. Both backends share the same process-agnostic and detector-agnostic API: users select a detector card—analogueous to choosing a detector card in DELPHES—and the same tool can be applied to arbitrary physics processes without re-training. Unlike C++/ROOT-based tools such as DELPHES, Parnassus runs natively on GPU (CUDA and Apple MPS) and requires no ROOT installation. The current release ships with CMS and ATLAS parametric cards validated against C++ DELPHES 3.5.0, and a CMS neural model trained on 2011 open data; additional detector models will be provided in future releases. We describe the installation, command-line and Python API, configuration system, and demonstrate the framework on Standard Model and BSM processes.

Keywords: fast simulation, detector simulation, particle flow, generative models, flow matching, diffusion models, differentiable simulation, high energy physics

PROGRAM SUMMARY

Program title: Parnassus

CPC Library link to program files:

(to be added by Technical Editor)

Developer’s repository link: <https://github.com/parnassus-hep/parnassus>

Licensing provisions: MIT

Programming language: Python (≥ 3.12)

Nature of problem: Full detector simulation and reconstruction based on GEANT4 is computationally expensive, often dominating the computing budget of large-scale High Energy Physics analyses. A fast, accurate surrogate is needed that maps truth-level (generator-level) particles directly to detector-level particle-flow objects, preserving the fidelity of the full chain while being orders of magnitude faster.

Solution method: Parnassus offers two interchangeable generator backends sharing a common API. The *neural backend* trains conditional flow matching models [7, 8] on full simulation and reconstruction data; the model is conditioned on truth-level particle properties and generates detector-level particle-flow

objects including kinematics, classification, and track impact parameters. The *parametric backend*, TORCHDELPHES, re-implements the DELPHES [25] fast-simulation chain (particle propagation, tracking efficiency, momentum smearing, calorimeter response, energy-flow merging) in pure PyTorch, validated against C++ DELPHES 3.5.0 for CMS and ATLAS cards. Both backends are followed by physics-based post-processing (jet clustering via FASTJET [17], lepton isolation). Users select a detector card—analogueous to a DELPHES card—to target a specific experiment.

Additional comments including restrictions and unusual features:

The framework is implemented entirely in Python with PyTorch and runs natively on CPU, CUDA GPU, and Apple MPS. It has no dependency on ROOT; output to ROOT format is optionally handled via `uproot` [24]. The current release ships with CMS and ATLAS parametric (TORCHDELPHES) cards and a CMS neural model trained on 2011 open data [6]; additional detector models for other experiments will be added in future releases. The neural CMS model supports up to 400 truth particles per event; events exceeding this limit retain the 400 highest- p_T particles.

1 Introduction

Parnassus is a **pure-Python, fully GPU-compatible** framework for **fast detector simulation and reconstruction** in High Energy Physics (HEP). It replaces the computationally expensive full GEANT4 [3] simulation and reconstruction chain with generative models that map truth-level (generator-level) particles directly to detector-level particle-flow [4, 5] (pflow) objects, bypassing intermediate steps such as hit simulation, digitisation, and track reconstruction. The framework is both **process-agnostic** and **detector-agnostic**: users select a detector card to target a specific experiment, and the same tool can be applied to arbitrary physics processes by simply providing different input truth events.

Parnassus exposes two interchangeable **generator backends** sharing a common API:

- A **neural backend** based on conditional flow matching [7, 8] trained on full simulation and reconstruction data, providing high-fidelity learned detector response.
- A **parametric backend**, TORCHDELPHES, which re-implements the DELPHES [25] fast-simulation chain (particle propagation, tracking efficiency, momentum smearing, calorimeter response, energy-flow merging) in pure PyTorch. The implementation is validated against C++ DELPHES 3.5.0 for both CMS and ATLAS detector cards and provides a fully differentiable, GPU-native drop-in alternative to running C++ DELPHES.

Unlike traditional fast simulation tools such as DELPHES [25], which are written in C++, require the ROOT framework [19], and run only on CPU, Parnassus is implemented entirely in Python with PyTorch [22] and runs natively on GPU (CUDA and Apple MPS), enabling seamless integration into modern Python-based analysis workflows and significant acceleration on GPU hardware. ROOT I/O is supported via `uproot` [24] for writing output files, but is entirely optional — Parnassus has no dependency on ROOT itself.

Users select a **detector card** — analogueous to choosing a detector card in DELPHES — to target a specific experiment. The current release ships with CMS and ATLAS TORCHDELPHES parametric cards, and a CMS neural model trained on 2011 open data [6]; additional detector models will be added in future releases.

The physics methodology underlying Parnassus has been described in detail in Refs. [1, 2], where we demonstrated that the approach reproduces full CMS simulation and reconstruction across a wide range of Standard Model and BSM processes. In this document we focus on the **public software release** — its installation, usage, and API — to enable the broader HEP community to adopt neural fast simulation and reconstruction in their workflows.

The neural backbone uses conditional flow matching [7, 8] to train continuous-time diffusion models [9, 10] that are sampled via Euler integration at inference time. The models are conditioned on truth-level particle properties, making the generation inherently process-agnostic without requiring retraining for new physics scenarios.

1.1 Design goals

Speed Orders of magnitude faster than full simulation and reconstruction by replacing the detector response with neural diffusion models. Fully GPU-compatible for additional acceleration.

Pure Python

Implemented entirely in Python with no compiled dependencies beyond PyTorch. Unlike C++/ROOT-based tools such as DELPHES [25], Parnassus integrates naturally into modern Python analysis ecosystems. ROOT output is available via `uproot` but is not required.

Fidelity Each detector model is trained on full simulation and reconstruction data to reproduce realistic detector-level distributions including particle kinematics, vertex positions, impact parameters, and particle identification. Excellent agreement with full simulation has been demonstrated across diverse SM and BSM processes [1, 2].

Flexibility Accept truth-level input from any source — PYTHIA8 [11] on-the-fly generation, HepMC [12] files from external generators (MADGRAPH5_AMC@NLO [13], SHERPA [14], HERWIG [15], POWHEG [16]), or custom formats. Users select a detector model to target a specific experiment, analogous to choosing a detector card in DELPHES.

Completeness

Full post-processing pipeline including jet clustering (FASTJET [17]) and lepton isolation, with optional output to ROOT format via `uproot` [24].

1.2 Output quantities

Given a set of truth-level particles for each event, Parnassus generates the quantities listed in Table 1.

Table 1: Output quantities produced by Parnassus for each event.

Output	Description
Particle-flow particles	Full kinematics (p_T , η , ϕ), vertex (v_x , v_y , v_z), class, PDG ID
Impact parameters	d_0 , z_0 and their errors for charged particles
Jets	Clustered with configurable algorithms (anti- k_t [18], gen- k_t); substructure (D_2 [20], C)
Lepton isolation	Isolation scores and p_T sums in configurable ΔR cones
Event-level quantities	H_T , $\vec{E}_T^{\text{miss}}$ components, particle multiplicity

2 Architecture overview

Parnassus passes truth-level events through a detector model followed by physics-based post-processing. The detector model can be either a neural backend or the TORCHDELPHES parametric backend; both share the same input and output interface so that downstream post-processing is unchanged. The pipeline is illustrated in Fig. 1.

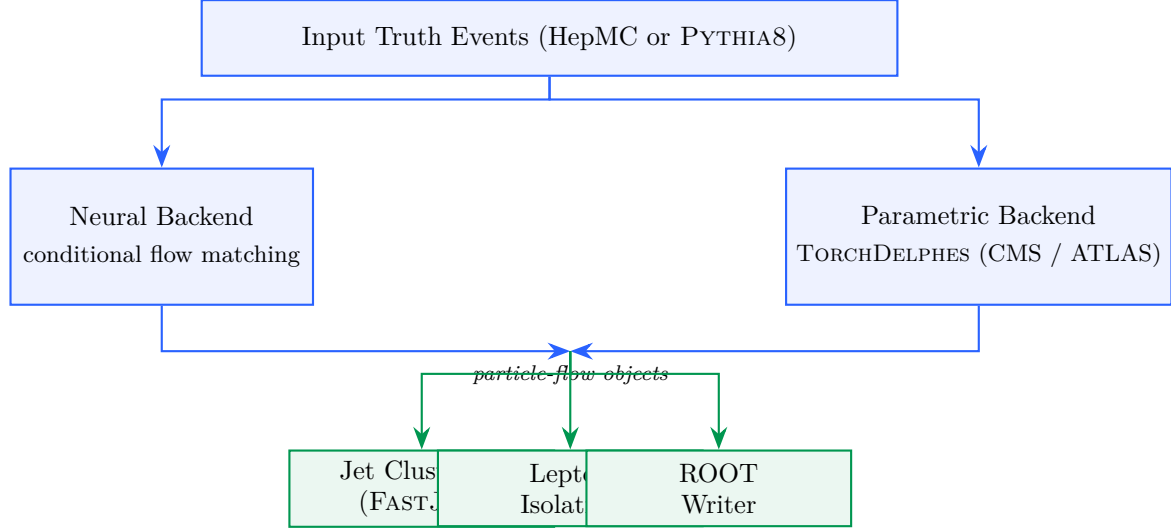


Figure 1: Parnassus generation pipeline. Truth-level events are routed through either the neural backend (conditional flow matching) or the parametric backend (TORCHDELPHES), both producing particle-flow objects. Physics-based post-processing (green) computes derived quantities and writes the final output.

The **neural backend** is a diffusion model trained with conditional flow matching [7, 8] and sampled via Euler integration. The model is conditioned on truth-level particle information, making the generation process inherently dependent on the input physics process without being tied to any specific one. The number of diffusion steps is configurable at runtime via the `--num_steps (-n)` flag, allowing users to trade generation speed for quality.

The **parametric backend**, TORCHDELPHES, is a pure-PyTorch re-implementation of the DELPHES [25] fast-simulation chain. It models particle propagation through the magnetic field, tracking efficiency, momentum smearing, calorimeter response, and energy-flow merging, with per-module parameters loaded from the selected detector card. The current release provides CMS and ATLAS cards ported from their respective DELPHES TCL cards and validated against C++ DELPHES 3.5.0. Being implemented in pure PyTorch, TORCHDELPHES runs natively on GPU and shares the same input and output conventions as the neural backend.

Both backends expose the same output contract (particle-flow kinematics, vertex positions, particle classification, and, where supported, track impact parameters), so downstream post-processing is identical.

3 Installation and quick start

3.1 Installation

```

# Clone the repository
git clone https://github.com/parnassus-hep/parnassus.git
cd parnassus

# Install with pip (requires Python 3.12)
pip install .

# For development
pip install -e ".[dev]"

```

Listing 1: Installation commands.

3.2 Requirements

- Python ≥ 3.12 , < 3.13
- PyTorch [22] $\geq 2.6.0$
- Pythia8mc [11] 8.316.0
- PyHepMC [12] $\geq 2.14.0$
- FastJet [17] $\geq 3.4.3.1$
- EnergyFlow [23] $\geq 1.4.0$
- Uproot [24] $\geq 5.5.2$ (*optional, for ROOT output*)

3.3 Initialise a configuration file

```
parnassus init .
```

This copies the default `default_config.yaml` to the current directory. The default configuration uses the `cms_2011_flow_v00` neural detector model with anti- k_t jet clustering and lepton isolation. To target a different detector or to switch to the parametric (TORCHDELPHES) backend, change the `generator.type` and `generator.name` fields in the YAML configuration. The two generator backends are configured as shown in Listing 2.

```
# --- Neural backend (conditional flow matching) ---
generator:
  type: "neural"
  name: "cms_2011_flow_v00" # trained CMS model
  num_steps: 50
  batch_size: 2000
  device: "cpu" # "cpu", "cuda", or "mps"

# --- Parametric backend (TorchDelphes) ---
generator:
  type: "parametric"
  name: "cms" # "cms" or "atlas" detector card
  seed: 42 # optional, for reproducibility
```

Listing 2: Switching between the neural and parametric (TORCHDELPHES) generator backends.

3.4 Run with a HepMC input file

```
parnassus run \
-c default_config.yaml \
-i events.hepmc \
-ne 1000 \
-bs 2000 \
-o output.root
```

Listing 3: Basic command-line invocation.

The available command-line flags are listed in Table 2.

Table 2: Command-line flags for `parnassus run`.

Flag	Long form	Description
-c	--config	Path to YAML configuration file
-i	--input_path	Input HepMC or Pythia <code>.cmnd</code> file
-o	--output_path	Output ROOT file path
-ne	--num_events	Number of events to process
-bs	--batch_size	Batch size for neural generation
-n	--num_steps	Diffusion sampler steps (default: 50)

3.5 Run with Pythia8 on-the-fly generation

To generate truth-level events on-the-fly with PYTHIA8, create a `.cmnd` steering card and pass it as the input file. Parnassus detects `.cmnd` files automatically:

```
parnassus run \
  -c default_config.yaml \
  -i ttbar.cmnd \
  -ne 5000 \
  -bs 2000 \
  -o ttbar_fastsim.root
```

4 Demonstration: SM and BSM processes

The neural models are **process-agnostic** — they learn the detector response as a function of truth-level particle properties, not the physics process itself. We have previously demonstrated [1, 2] that Parnassus reproduces full simulation and reconstruction across a wide range of SM and BSM processes. Here we show two representative examples using the CMS detector model to illustrate the API in action: a Standard Model process ($H \rightarrow ZZ \rightarrow 4\ell$) and a BSM exotic Higgs decay ($H \rightarrow aa \rightarrow gg\mu\mu$). The same workflow applies to any detector model by changing the `generator.name` in the configuration.

The current CMS detector model supports up to 400 truth particles per event. For events exceeding this limit, Parnassus retains the 400 highest- p_T particles so that no events are discarded.

4.1 Standard Model: $H \rightarrow ZZ \rightarrow 4\ell$

We process 100 events from a HepMC input file produced by PYTHIA8 [11]:

```
parnassus run \
  -c default_config.yaml \
  -i h4l_events.hepmc \
  -ne 100 \
  -o h4l_output.root
```

4.2 BSM: $H \rightarrow aa \rightarrow gg\mu\mu$

For the BSM example, we generate exotic Higgs decays on-the-fly with PYTHIA8. The Higgs decays to a pair of light pseudoscalars a ($m_a = 20$ GeV), each decaying to either gg or $\mu^+\mu^-$, producing a mixed final state of jets and muon pairs:

```

229 ! haa_ggmumu.cmd -- BSM exotic Higgs decay
230 Beams:eCM = 13000.
231 Beams:idA = 2212
232 Beams:idB = 2212
233
234 ! BSM Higgs production via gluon fusion
235 Higgs:useBSM = on
236 HiggsBSM:gg2H2 = on
237 35:m0 = 125.0 ! Higgs mass
238
239 ! SM-like couplings for production
240 HiggsH2:coup2u = 1.0
241 HiggsH2:coup2d = 1.0
242 HiggsH2:coup2A3A3 = 1.0 ! H -> aa coupling
243
244 ! Force H -> a a only
245 35:onMode = off
246 35:onIfAll = 36 36
247
248 ! Light pseudoscalar a properties (PDG ID 36)
249 36:m0 = 20.0 ! mass = 20 GeV
250 HiggsA3:coup2d = 1.0
251 HiggsA3:coup2u = 1.0
252 HiggsA3:coup2l = 1.0
253
254 ! Force a -> gg or mumu only
255 36:onMode = off
256 36:onIfAny = 21 13
257
258

```

Listing 4: Pythia8 steering card for BSM $H \rightarrow aa \rightarrow gg\mu\mu$.

```

259 parnassus run \
260   -c default_config.yaml \
261   -i haa_ggmumu.cmd \
262   -ne 1000 \
263   -o haa_ggmumu_output.root
264
265

```

This demonstrates running Parnassus on a BSM signal process not present in the training data. The mixed $gg\mu\mu$ final state probes the model’s ability to simultaneously generate hadronic jets and isolated muons within the same event.

4.3 Kinematic distributions

Fig. 2 compares the particle-flow kinematic distributions for both processes. The p_T spectra follow the expected steeply falling shape, the η distributions reflect the detector acceptance, and the multiplicity distributions reflect the different underlying event structures. The BSM $H \rightarrow aa$ process shows higher multiplicity due to the hadronic activity from the $a \rightarrow gg$ decays.

4.4 Event display

Fig. 3 shows a representative $H \rightarrow 4\ell$ event in the η - ϕ plane, illustrating how Parnassus transforms truth-level particles into detector-level pflow objects.

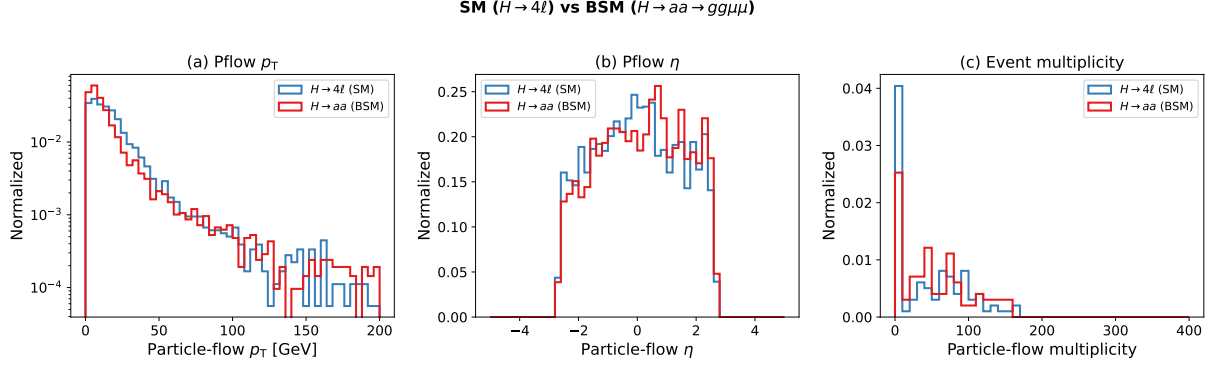


Figure 2: Particle-flow kinematic distributions comparing SM ($H \rightarrow 4\ell$, 99 events) and BSM ($H \rightarrow aa \rightarrow gg\mu\mu$, 99 events) processes. (a) p_T in log scale, (b) pseudorapidity η , (c) particle-flow multiplicity per event. All distributions are normalised to unit area.

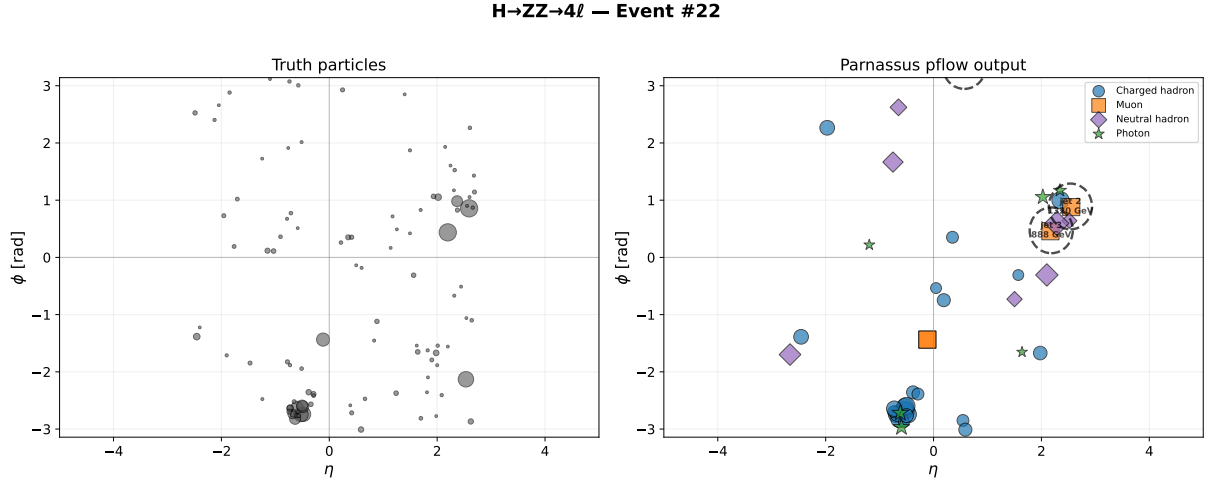


Figure 3: Event display for $H \rightarrow ZZ \rightarrow 4\ell$: truth particles (left) and Parnassus pflow output (right) in the η - ϕ plane. Marker size scales with p_T ; dashed circles indicate anti- k_t jets ($R = 0.5$). Particle classes: charged hadrons (blue circles), electrons (red triangles), muons (orange squares), neutral hadrons (purple diamonds), photons (green stars).

4.5 Aggregate statistics

Table 3 compares the aggregate statistics for both processes. The BSM process shows higher truth-particle multiplicity (163 ± 70) due to the $a \rightarrow gg$ hadronic decays, while the pflow output multiplicity is comparable across both processes.

Table 3: Aggregate statistics for SM and BSM processes (mean $\pm$ std).

Quantity	$H \rightarrow 4\ell$ (99 evt)	$H \rightarrow aa$ (99 evt)
Truth particles/event	97.2 ± 46.0	163.4 ± 69.9
Pflow particles/event	46.3 ± 47.2	53.6 ± 46.4

4.6 Particle class distribution

Fig. 4 compares the particle class composition between the two processes. Both show similar class fractions dominated by charged hadrons ($\sim 60\%$), consistent with the process-agnostic nature of the model.

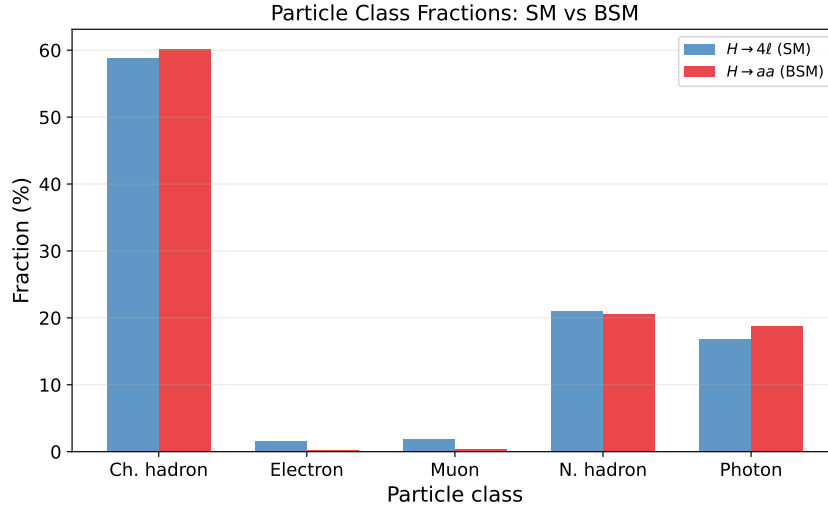


Figure 4: Particle class fractions for SM ($H \rightarrow 4\ell$) and BSM ($H \rightarrow aa \rightarrow gg\mu\mu$) processes. The model produces consistent class compositions regardless of the input physics process.

5 Diffusion steps convergence

The number of diffusion steps N controls the trade-off between generation quality and speed. Fig. 5 shows the convergence of particle-flow kinematic distributions as a function of N for the $H \rightarrow 4\ell$ sample.

Fig. 6 shows that particle class fractions are also stable across step counts.

Table 4 summarises the generation time as a function of diffusion steps on CPU.

The default of 50 steps provides a good balance of quality and speed. For quick validation runs, 25 steps yields nearly identical distributions at lower cost.

6 Python API

For programmatic usage, Parnassus provides a clean Python API:

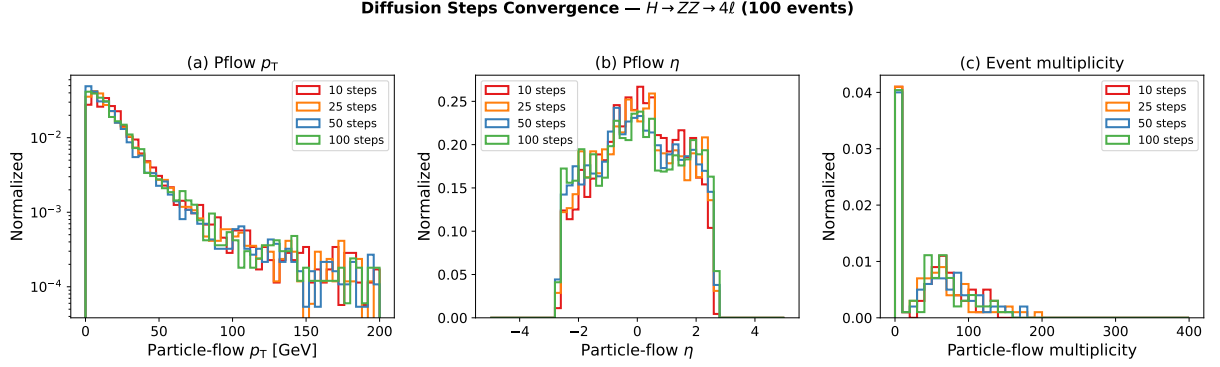


Figure 5: Diffusion steps convergence for $H \rightarrow ZZ \rightarrow 4\ell$ (100 events). (a) Pflow p_T , (b) pflow η , (c) event multiplicity. Distributions are stable from $N = 25$ onward.

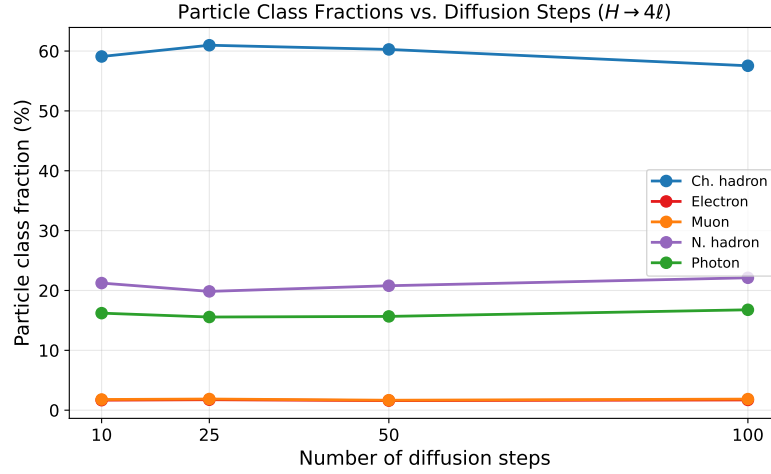


Figure 6: Particle class fractions vs. number of diffusion steps. All classes converge by $N = 25$.

Table 4: Generation time vs. diffusion steps (100 events, CPU).

Steps	Wall time
10	~1:30
25	~1:30
50	~2:10
100	~5:20

```

295
296 from pathlib import Path
297 from parnassus.configs import Config
298 from parnassus.pipelines import (
299     GenerationPipeline,
300     JetClusteringPipeline,
301     IsolationPipeline,
302 )
303 from parnassus.configs.pipeline import (
304     JetClusteringConfig,
305     IsolationConfig,
306 )
307 from parnassus.writers import RootWriter
308
309 # Load configuration
310 config = Config.from_yaml("my_config.yaml")
311
312 # Override settings programmatically
313 config.dataset_config.file_path = (
314     Path("my_events.hepmc").absolute()
315 )
316 config.dataset_config.num_events = 5000
317
318 # Run generation
319 pipeline = GenerationPipeline(config)
320 gen_events, accessors_dict = pipeline.run()
321
322 # Post-processing
323 for pipeline_config in config.pipeline_configs:
324     if isinstance(pipeline_config, JetClusteringConfig):
325         jet_pipeline = JetClusteringPipeline(pipeline_config)
326         jet_pipeline.process(gen_events)
327     elif isinstance(pipeline_config, IsolationConfig):
328         iso_pipeline = IsolationPipeline(pipeline_config)
329         iso_pipeline.process(gen_events)
330
331 # Inspect generated data
332 for event in gen_events[:5]:
333     print(f"Event {event.event_number}:")
334     n_truth = len(event.truth_particles.pt)
335     n_pflow = len(event.pflow_particles.pt)
336     print(f"  Truth particles: {n_truth}")
337     print(f"  Pflow particles: {n_pflow}")
338     print(f"  HT = {event.ht:.1f} GeV")
339
340 # Write to ROOT
341 writer = RootWriter(config.writer_config)
342 writer.write(gen_events)
343

```

Listing 5: Programmatic usage of the generation pipeline.

344 6.1 Reading output files

```

345
346 import uproot
347 import numpy as np
348
349 f = uproot.open("output.root")

```

```

350 tree = f["Parnassus"]
351
352 # Access pflow jet pT
353 jet_pt = tree["PflowJetsAntiKt05.pt"].array()
354
355 # Access electron isolation
356 e_iso = tree["Electrons.iso_var"].array()
357
358 # Access impact parameters
359 d0 = tree["Pflow.d0"].array()
360 z0 = tree["Pflow.z0"].array()
361

```

Listing 6: Reading the output ROOT file with uproot.

7 Configuration reference

```

363 # -- Dataset -----
364 dataset:
365     file_path: ""           # Path to .hepmc or .cmnd file
366     num_events: 1000       # Number of events to process
367     entry_start: 0         # Starting event index (HepMC)
368
369 # -- Generator (choose one backend) -----
370 # Option A: Neural backend (conditional flow matching)
371 generator:
372     type: "neural"         # Select the neural backend
373     name: "cms_2011_flow_v00" # Detector model (like a Delphes card)
374     num_steps: 50          # Number of flow-matching integration
375                             steps
376     batch_size: 2000       # Events per batch
377     device: "cpu"          # "cpu", "cuda", or "mps"
378
379 # Option B: Parametric backend (TorchDelphes)
380 # generator:
381 #     type: "parametric"    # Select the TorchDelphes backend
382 #     name: "cms"           # Detector card: "cms" or "atlas"
383 #     max_particles: 10000  # Maximum particles per event
384 #     seed: 42              # Random seed for stochastic smearing
385
386 # -- Post-processing Pipelines -----
387 pipelines:
388     TruthJetsAntiKt05:
389         type: "cluster"
390         collection: truth
391         dr: 0.5
392         algorithm: antikt
393         pt_min: 10
394         nconst_min: 2
395     PflowJetsAntiKt05:
396         type: "cluster"
397         collection: pflow
398         dr: 0.5
399         algorithm: antikt
400         pt_min: 10
401         nconst_min: 2
402     ElectronIsolation:
403

```

```

404     type: "isolation"
405     collection: "electrons"
406     dr: 0.4
407 MuonIsolation:
408     type: "isolation"
409     collection: "muons"
410     dr: 0.4
411
412 # -- Output -----
413 output:
414     file_path: ""           # Output ROOT file path
415     format: default
416

```

Listing 7: Annotated default configuration.

8 Conclusion

We have presented the public release of Parnassus, a pure-Python, GPU-compatible framework for fast simulation and reconstruction in High Energy Physics. Unlike traditional C++/ROOT-based tools such as DELPHES [25], Parnassus runs entirely in Python with PyTorch, supports GPU acceleration (CUDA and Apple MPS), and requires no ROOT installation — ROOT output is optionally handled via `uproot`. The software provides a simple command-line and Python API for generating detector-level particle-flow objects from truth-level input, with full post-processing including jet clustering and lepton isolation.

Parnassus offers two interchangeable generator backends that share a common user-facing API: a **neural backend** based on conditional flow matching models trained on full simulation, and a **parametric backend** (TORCHDELPHES) that re-implements the DELPHES fast simulation chain — particle propagation, tracking efficiency, momentum smearing, calorimetry, and energy-flow merging — as differentiable PyTorch modules. TorchDelphes has been validated against C++ DELPHES 3.5.0 and provides a GPU-accelerated, differentiable drop-in replacement that is useful both as a standalone fast simulator and as a baseline for the neural backend.

The framework is both process-agnostic and detector-agnostic. Users select a detector model — analogous to choosing a detector card in DELPHES — to target a specific experiment. The current release ships with CMS and ATLAS parametric cards for the TorchDelphes backend, and a CMS neural model validated against full CMS simulation across diverse SM and BSM processes [1, 2]; additional detector models and neural checkpoints will be provided in future releases.

The source code is publicly available at <https://github.com/parnassus-hep/parnassus> under the MIT license.

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